

ENGE702 Engineering Mathematics III

Tutorial 1

Written Part

1. Solve the following ODEs using integration:

(a) $y' = 3x^2 + 1$ $y = x^3 + x + c$

(c) $y'' + 2x = 1$

(b) $y' - \cos(x) = \frac{1}{x}$ $y = \sin(x) + \ln(x) + c$

(d) $y'' + 3x^3 - 2x = x^2$ $y = -\frac{3x^5}{20} + \frac{x^4}{12} + \frac{x^3}{3} + cx + k$

2. Solve the following initial value problems:

(a) $y' = 3x^2 - 1$, $y(0) = 1$.

$y = x^3 - x + 1$

(b) $y' - \cos(2x) = 1$, $y(\pi) = 2$.

$y = \frac{1}{2}\sin(2x) + x + (2 - \pi)$

(c) $y' = 3e^{2x}$, $y(1) = \pi$.

$y = \frac{3}{2}e^{2x} + \pi - \frac{3}{2}e^2$

3. Solve the following initial value problems, given their general solutions:

(a) $y' + 2xy = 0$, $y(0) = 1$, general solution $y = ce^{-x^2}$

$y = e^{-x^2}$

(b) $x^2y' + 2xy = 3x + 1$, $y(1) = 0$, general solution $y = \frac{c}{x^2} + \frac{1}{x} + \frac{3}{2}$

$y = -\frac{5}{2x^2} + \frac{1}{x} + \frac{3}{2}$

(c) $y'' + 3y' + 2y = 0$, $y(0) = 1$, $y'(0) = 0$, general solution $y = c_1e^{-2x} + c_2e^{-x}$

$y = -e^{-2x} + 2e^{-x}$

4. (a) Verify that

$$u(x, t) = (Ax + B)(Ct + D) + (E \sin(\kappa x) + F \cos(\kappa x))(G \sin(\kappa ct) + H \cos(\kappa ct))$$

is a solution for the one dimensional wave equation

(find partial derivatives and show the equality holds)

$$c^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$$

for any constants $A, B, C, D, E, F, G, H, \kappa$.

5. Show the following functions are solutions to $y''' - 3y'' + 3y' - y = 0$:

(a) $y = e^t$

(c) $y = t^2 e^t$

(find derivatives and show the equation holds)

(b) $y = te^t$

(d) $y = ae^t + bte^t + ct^2e^t$

for any constants a, b , and c .

6. For what values of the constant λ is $y = e^{\lambda t}$ a solution to the following differential equations:

(a) $y' + 3y = 0$ $\lambda = -3$

(c) $y'' - 2y' + y = 0$ $\lambda = 1$

(b) $y' + 3y^2 = 0$ No values of λ

(d) $y'''' - 6y'' + 5y = 0$ $\lambda = \pm 1, \pm \sqrt{5}$

7. (a) Show that $y = e^{0.4t}$ is a solution to the differential equation $y' = 0.4y$.
(b) Show that $y = e^{kt}$ is a solution to the differential equation $y' = ky$.
(c) Write a short paragraph explaining the shape of the graph of the solution and the role k plays (particularly if k is positive or negative). Also explain how an initial condition affects the shape of the graph (particularly if the initial value is positive or negative).
- } find derivatives, show equation holds.
It is always exponential, bigger k makes it 'steeper' initial condition gives initial height.

8. Finished all these? Work through problem sets 1.1 and 1.2 in the textbook

Computer Part

In this session we will introduce matlab, and also the resources available to help you to learn to use matlab.

AUT has a licence with the matlab creators that allows you to access an array of online training material, and also download the latest version of matlab to your personal computer (if you wish - you don't need to do this for this course if you don't want to, you can access matlab on the computers at AUT).

To access these resources you will need to create an account with mathworks (the company that produces matlab).

You can do this by visiting [mathworks.com](https://www.mathworks.com) (or google 'matlab'). You can create an account by clicking the 'login' icon on the top right. **It is important you use your AUT email address to create the account, or you will not be able to access the material.**

From your account page, click 'Self-Paced Courses'.

If you have never used matlab before, it would be good to catch up with the basics. Work through the course 'MATLAB Onramp'. This may take some time (and will need to be completed in your own time), but will be helpful background material.

If you have at least a basic level of matlab experience, start working through sections 1 and 2 of the module 'Solving Ordinary Differential Equations with MATLAB'. We will continue with these sections next week.